Task 3: School Transport Management

BCS | Java | 40 minutes | Instructor: Dr Muhammad Shahid Bhatti

Implement a familiar transport scenario from the requirements.

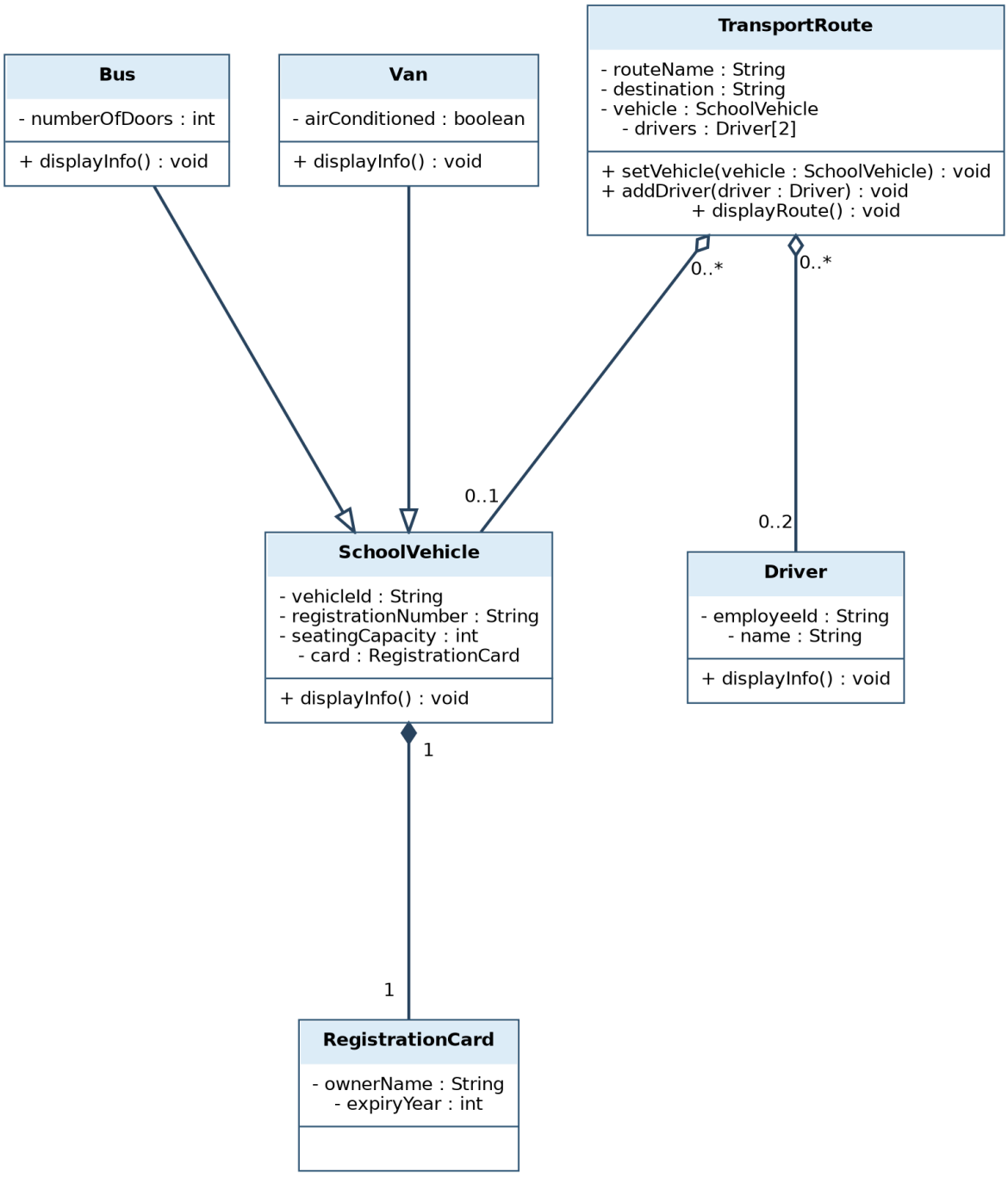
Scenario

A school uses vehicles to bring students to campus. Each SchoolVehicle has a vehicle ID, registration number, and seating capacity. A Bus additionally stores its number of doors. A Van additionally stores whether air conditioning is available.

Each vehicle has a RegistrationCard with an expiry year and an owner name. The card is created as part of constructing its vehicle; this task treats it as owned by that vehicle.

A TransportRoute has a route name and destination. It uses one already-created school vehicle and up to two already-created Driver objects. A Driver has an employee ID and name. Drivers can be assigned to other routes later, so they are not created inside the route.

UML Class Diagram



Read the UML notation: hollow triangle = inheritance; black diamond = composition; white diamond = aggregation; plain line = association. Multiplicities show how many objects participate. The editable PlantUML source is provided separately.

Classes and Attributes

Class	Required attributes
SchoolVehicle	vehicleId, registrationNumber, seatingCapacity, RegistrationCard card
Bus extends SchoolVehicle	numberOfDoors
Van extends SchoolVehicle	airConditioned

RegistrationCard	ownerName, expiryYear
Driver	employeeId, name
TransportRoute	routeName, destination, SchoolVehicle vehicle, Driver[] drivers

Required Methods

- SchoolVehicle.displayInfo(); override it in Bus and Van.
- TransportRoute.setVehicle(SchoolVehicle vehicle) to attach an existing vehicle.
- TransportRoute.addDriver(Driver driver) to store up to two drivers.
- TransportRoute.displayRoute() to print the route, vehicle, and drivers.

What to Create in main()

- Create a Bus, a Van, and two Driver objects.
- Create two TransportRoute objects: Morning Route and Afternoon Route.
- Assign the Bus to Morning Route and the Van to Afternoon Route.
- Add drivers to the routes and print both route summaries.

Relationships to Demonstrate

Inheritance: Bus and Van extend SchoolVehicle. Composition: SchoolVehicle creates RegistrationCard. Aggregation: TransportRoute refers to existing vehicle and Driver objects.

Completion Checklist

- ☐ All classes compile and run.
- ☐ Constructors use super(...) in subclasses.
- ☐ Inherited display methods are overridden where specified.
- ☐ Composition objects are constructed within their owner.
- ☐ Aggregation objects are created first and added/passed afterward.
- ☐ The program runs and prints understandable output.

Submission

Save all .java source files for this task in its own folder. At the end of the lab, submit ONE ZIP file containing completed folders for Tasks 1–4. Do not submit four separate ZIP files.